

MADE IN EU MINED ELSEWHERE GALLIUM AND THE REALITY OF EU CHIP SOVEREIGNTY

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02 BACKGROUND + MOTIVATION + RESEARCH QUESTION

Gallium (Ga) is a semiconductor metal, that is primarily used in electronics and magnet production. EU categorises it as a critical raw material (CRM) due to its economic importance and the strong import dependency EU has on China (97% of Ga import to EU). Gallium is necessary for the green transition and digitalization due to its widespread use in integrated circuits (ICs) and Light-Emitting Diodes (LEDs)[6][7][13][14].

EU has through the CRM Act (CRMA) set goals for self-reliance on materials like Ga, while also setting goals in the EU Digital Decade and Chips Act for the increased demand and domestic production of semiconductors (ICs and LEDs amongst others)[6][7][14]. These two strategies pull in the same direction, requiring more of the EU domestic production, but how do they add up and what is the combined effect?

Research question: How do EU strategies for Critical Raw Materials and production of semiconductors reshape the EU gallium system, focusing on imported electronics semiconductor components products?

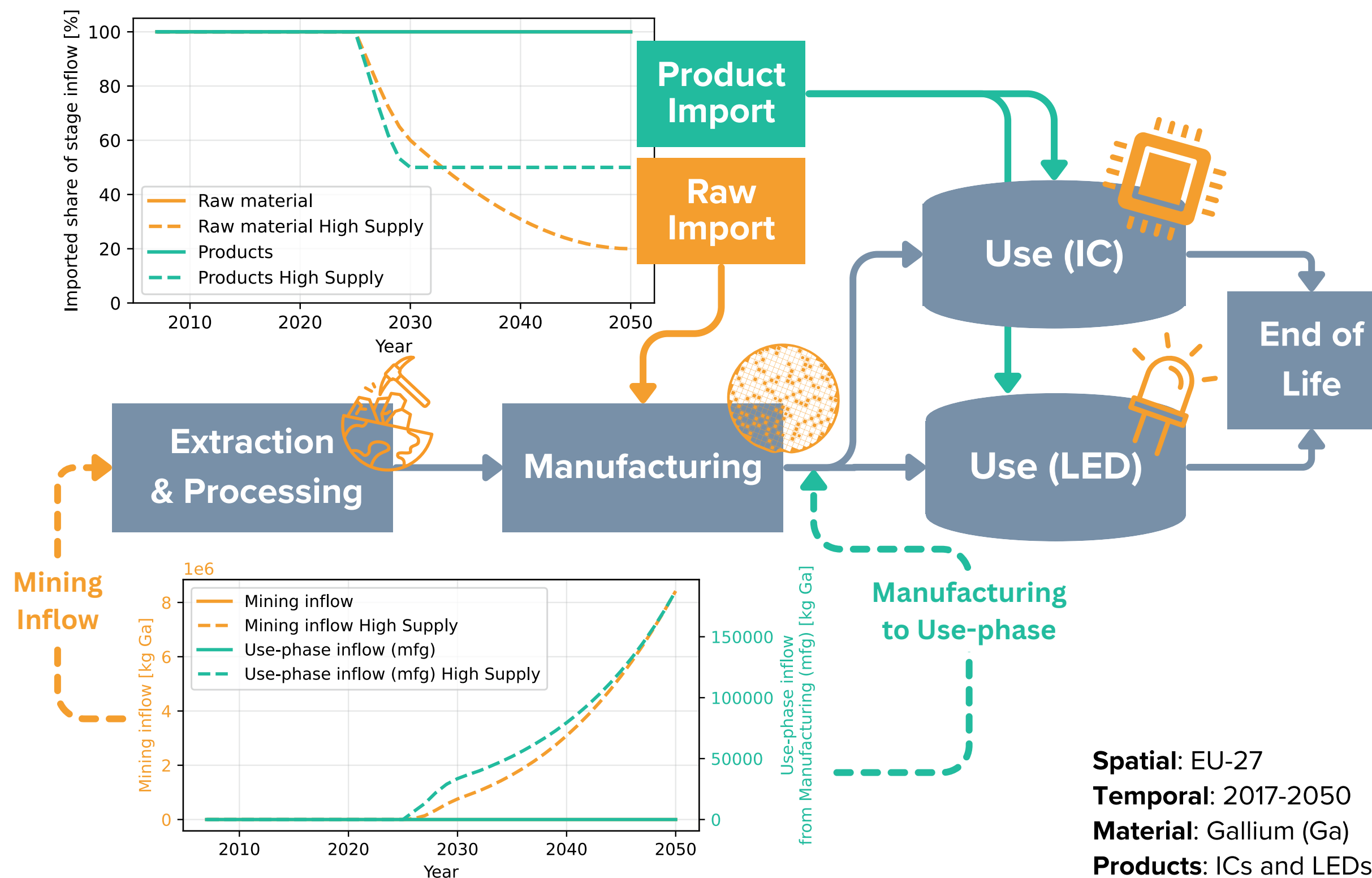
- How do import dependencies change when implementing EU strategies for self-sufficiency?

06 RESULTS + DISCUSSION + LIMITATIONS

Results and discussion: The combined effect of decreased import reliance and increased demand is considerable. As can be seen from the static snapshots, the loss from mining and manufacturing have a large impact on the extraction needed to satisfy the use-phase demand. Also even with the CRM fully implemented in 2030 (see **07**) the manufacturing and use-phase is still strongly dependent on import.

Research gap: Focusing on the EU, rather than than global [2][4][12][16][17][18] or regional models[4][8]. Using a scenario approach to policy evaluation, rather than market-based system dynamics[17].

Limitation and future research: Considers only historical import and not historical domestic production, and investigates only electronics components and not components in finished products. This is however aligned with and answers the RQ, but expanding the product selection should be considered for future research.



01 SUMMARY - TLDR

Gallium is a critical raw materials that is an essential part of electronics production of microchips (ICs) and LEDs. EU is currently highly dependent on import of both raw gallium and gallium containing electronics products and components. **This study answers:** How does EU policies for self-sufficiency and digitalization reshape the Ga system based on previously imported electronics components? **The results** show that localizing production relocates the upstream from import to domestic, and that added demand for domestic raw material from CRMA sufficiency goals and for electronics products for digitalization add up to require a large EU domestic extraction, processing, and manufacturing sector.

03 MODEL + METHODS

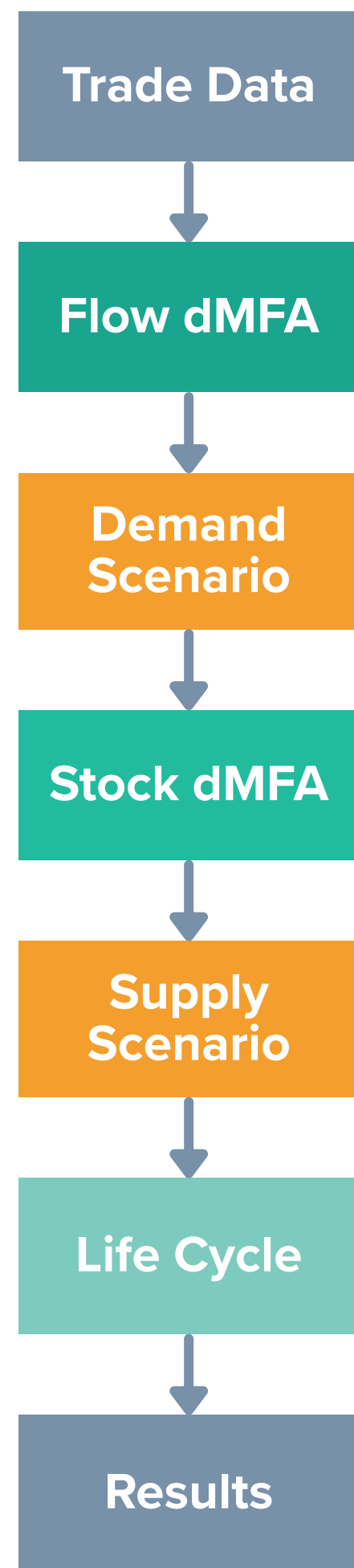
The model is built bottom-up using data from the EUROSTAT COMEXT database about EU net import of gallium containing electronics components [5].

Data about material intensities, market share, and survival curves for each product is derived from literature (Gao, 2024; Song, 2022).

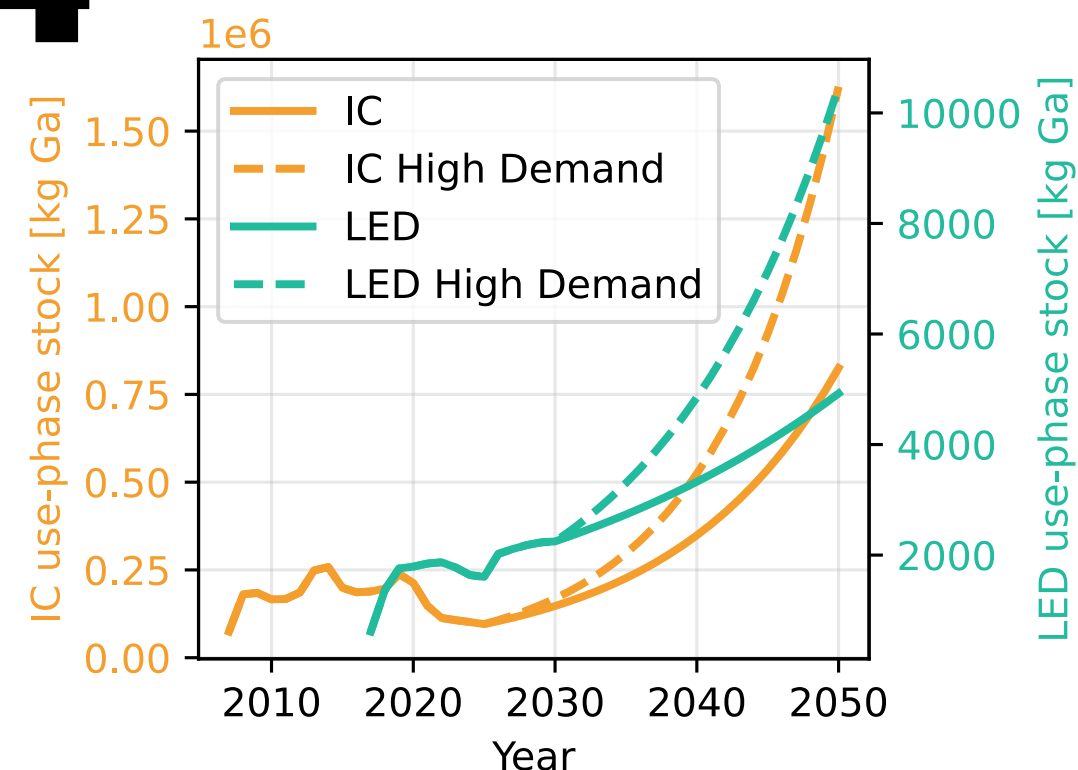
The trade and product data drives a flow-driven dynamic MFA (dMFA) that models the historical stock of the products.

Demand scenarios are then applied to create future stock data and fed into the stock-driven dMFA (see **04**)

Different supply scenarios are then applied through adjusting import vs. production share (see **05**). This together with Transfer-coefficients from literature (Løvik, 2015, 2016; Han, 2024; Ueberschaar, 2017) drive the calculation of flows through the different life-cycles stages, producing the final results



04 STOCK DEVELOPMENT

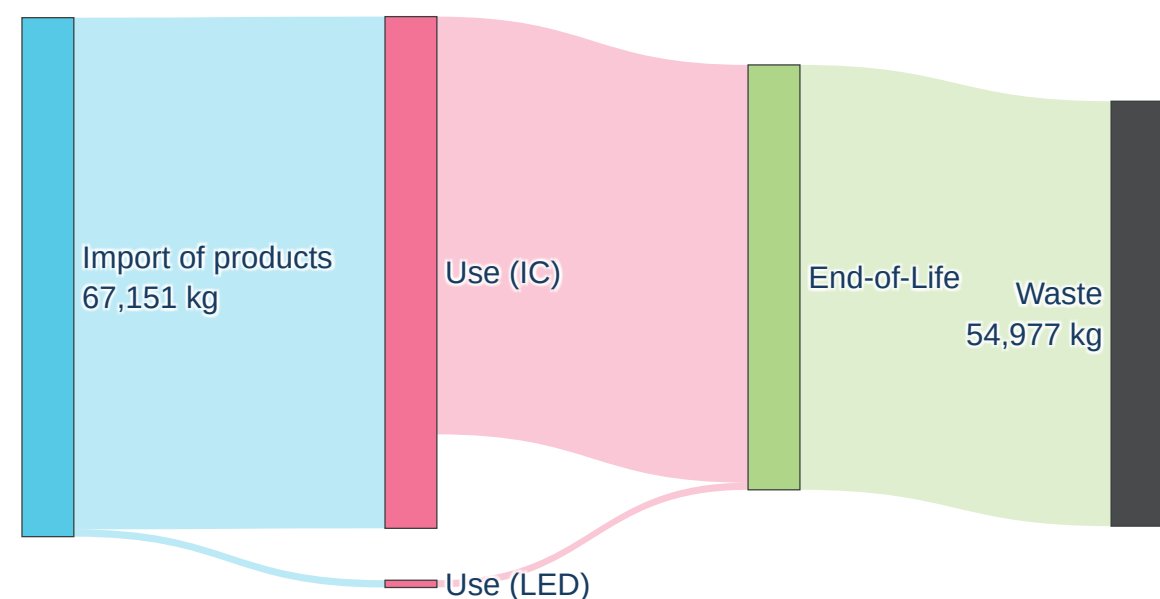


Where LEDs are expected to saturate around 2030 and then only slowly develop from there on [19], the expected future demand for ICs is exponential [6]. Each stock scenario is then further amplified in the high demand scenario.

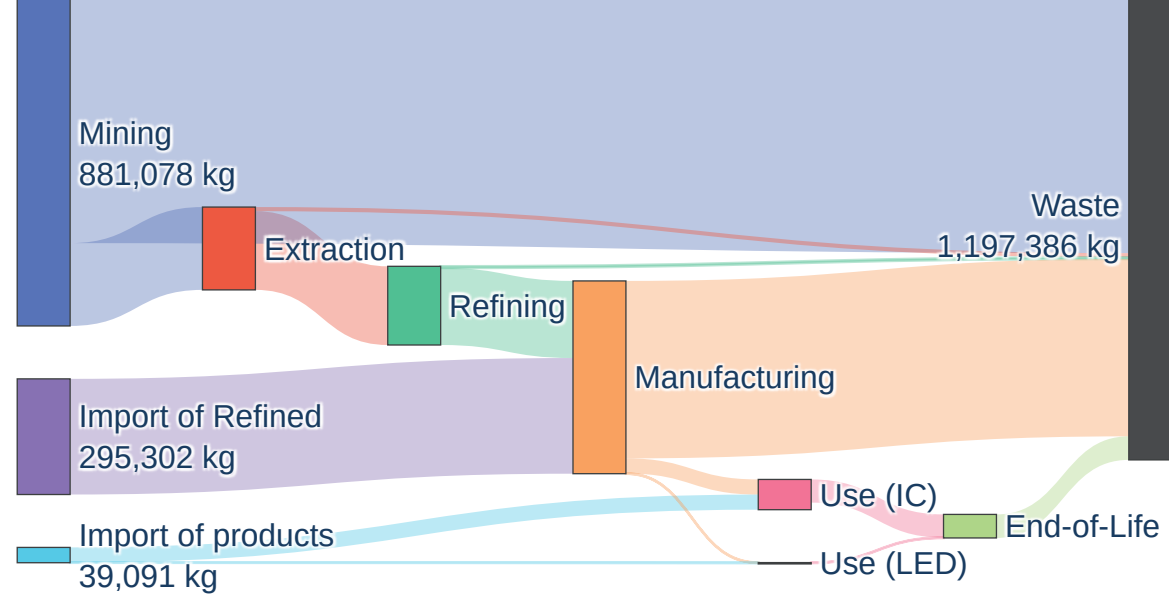
07 STATIC SNAPSHOTS

The static snapshots show the end of the EU-CRM and EU-DD strategies[1][7]. The difference in amount of input material for the EU is quite substantial. The mining required for the added manufacturing is more than 10x the status quo import products, while the import of products has not even halved.

STATUS QUO - 2030

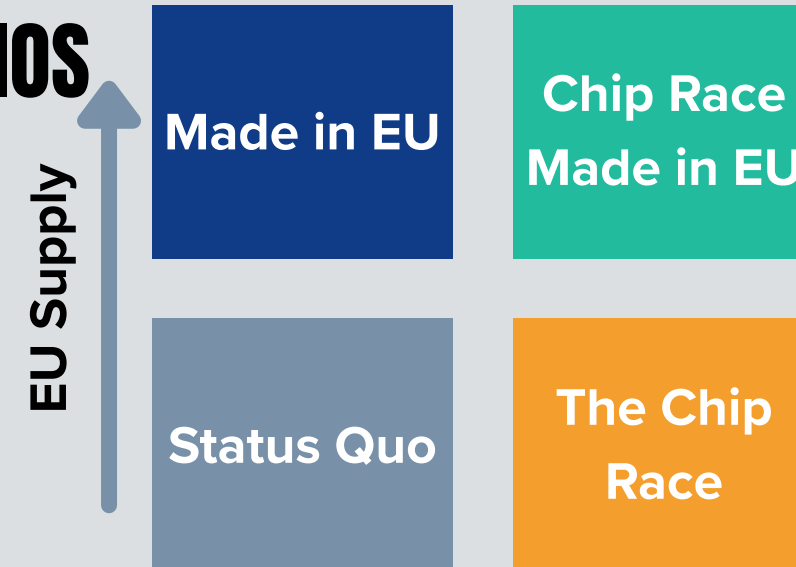


CHIP RACE MADE IN EU - 2030



05 SCENARIOS

The scenarios are based on the EU strategies for Critical Raw Material Act [7] and the Digital Decade Strategy [1]. These define both a higher demand for digital technology, as well as both raw material and product self-sufficiency.



The supply axis defines raw material and product domestic production. The demand axis defines increased use demand. If both policies are to be fulfilled, EU should land in the scenario "Chip Race, Made in EU"

00 REFERENCES

This list includes core references. For "[1]" references and a complete list, follow the link/QR-code:

- Gao, Z., Geng, Y., Li, M., Liang, J.-J., & Houssini, K. (2024). Tracking the global anthropogenic gallium cycle during 2000–2020: A trade-linked multiregional material flow analysis. Han, Z., Liu, Q., Ouyang, X., Song, H., Gao, T., Liu, Y., Wen, B., & Dai, T. (2024). Tracking two decades of global gallium stocks and flows: A dynamic material flow analysis. Jia, H., Zhou, Y., Wang, A., Wang, G., Li, T., Wang, C., Xing, W., Ma, Z., & Li, P. (2022). Evolution of the Anthropogenic Gallium Cycle in China From 2005 to 2020. Licht, C., Peiró, L. T., & Villalba, G. (2015). Global Substance Flow Analysis of Gallium, Germanium, and Indium: Quantification of Extraction, Uses, and Dissipative Losses within their Anthropogenic Cycles. Lovik, A. N., Restrepo, E., & Müller, D. B. (2015). The Global Anthropogenic Gallium System: Determinants of Demand, Supply and Efficiency Improvements. Lovik, A. N., Restrepo, E., & Müller, D. B. (2016). Byproduct Metal Availability Constrained by Dynamics of Carrier Metal Cycle: The Gallium–Aluminum Example. Song, H., Wang, C., Sen, B., & Liu, G. (2022). China Factor: Exploring the Byproduct and Host Metal Dynamics for Gallium–Aluminum in a Global Green Transition. Ueberschaar, M., Otto, S. J., & Rotter, V. S. (2017). Challenges for critical raw material recovery from WEEE – The case study of gallium.



Detailed and complete reference list can be found at the QR code or link below:
<http://bit.ly/4oLc7kH>

